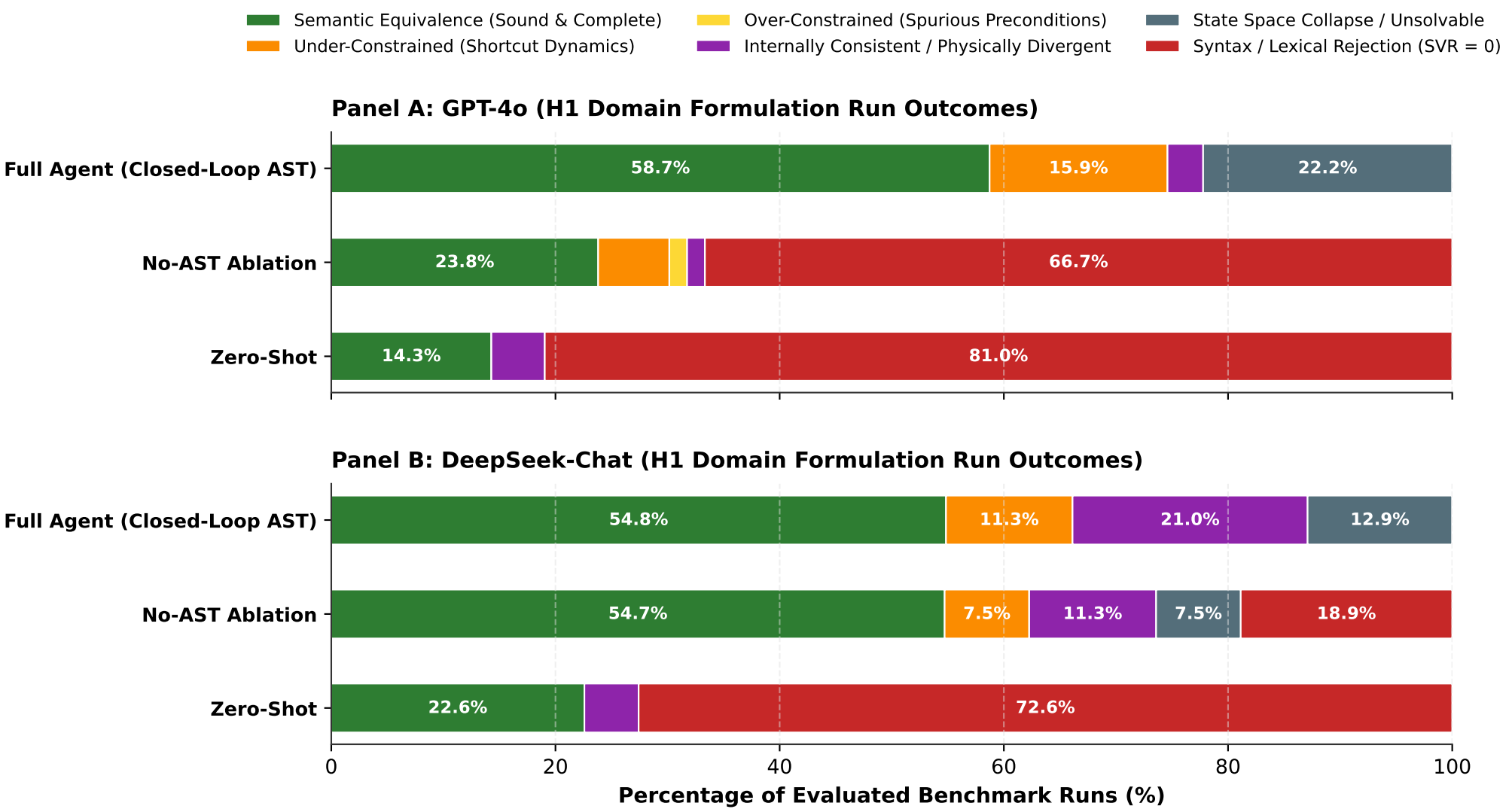
Semantic Equivalence (Sound & Complete)
Under-Constrained (Shortcut Dynamics)
Over-Constrained (Spurious Preconditions)
Internally Consistent / Physically Divergent
State Space Collapse / Unsolvable
Syntax / Lexical Rejection (SVR = 0)
Panel A: GPT-4o (H1 Domain Formulation Run Outcomes)
Full Agent (Closed-Loop AST)
58.7%
15.9%
22.2%
No-AST Ablation
23.8%
66.7%
Zero-Shot
14.3%
81.0%
Panel B: DeepSeek-Chat (H1 Domain Formulation Run Outcomes)
Full Agent (Closed-Loop AST)
54.8%
11.3%
21.0%
12.9%
No-AST Ablation
54.7%
7.5%
11.3%
7.5%
18.9%
Zero-Shot
22.6%
72.6%
0
20
40
60
80
100
Percentage of Evaluated Benchmark Runs (%)